12TH CHEMISTRY · CHAPTER 22 · FULL CHAPTER NOTES

Hydroxy Compounds

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] What are hydroxy compounds? Why are alcohols and phenols important?

Alcohols and phenols are hydroxy organic compounds with at least one hydroxyl group attached to an aliphatic or an aryl carbon, respectively. They are important because of their widespread occurrence and diverse applications. Ethanol (ethyl alcohol or grain alcohol) is produced from bio-based materials and is used as a biofuel. Phenol is an ingredient in disinfectants, such as soaps and hospital disinfectant liquids.

Q.2 [Short] How was phenol used in surgery after 1867? Why was it later replaced?

After Louis Pasteur's research revealed the existence of bacteria, scientists started cleaning surgical instruments and wounds with phenol solutions in 1867. A patient's chances of surviving surgery increased significantly because of the potent antiseptic properties of phenol solution. Since phenol burns healthy tissues, it was too potent for this use, and alternatives were eventually discovered.

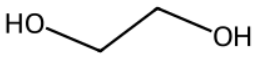
22.1 ALCOHOLS

Q.3 [Short] What are alcohols? Give their general formula and first two members.

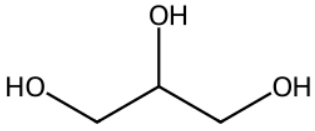
The hydroxyl group (–OH) is the functional group of all alcohols. Alcohols are represented by the general formula ROH, where R is an alkyl or aryl group. The family also includes familiar substances such as cholesterol and the carbohydrates. Methanol (CH_3OH) and ethanol (CH_3CH_2OH) are the first two members of the homologous series of monohydric alcohols.

Q.4 [Short] Differentiate between monohydric and polyhydric alcohols with examples.

Depending on the number of hydroxyl groups, alcohols are classified into **monohydric** alcohols (one –OH group), e.g., ethanol CH_3CH_2OH and propan-1-ol $CH_3CH_2CH_2OH$, and **polyhydric** alcohols (more than one –OH group), e.g., 1,2-ethanediol (glycol) and 1,2,3-propanetriol (glycerol).



1,2–ethanediol (glycol)



1,2,3–propanetriol (glycerol)

Q.5 [Long] Classify monohydric alcohols into primary, secondary and tertiary alcohols.

Depending on the number of carbon atoms directly attached to the carbon bonded to the –OH group:

Type	–OH group is attached to	General formula	Example
Primary (1°) alcohol	a primary carbon atom	R_1CH_2OH	ethanol, propan-1-ol
Secondary (2°) alcohol	a secondary carbon atom	R_2CHOH	propan-2-ol
Tertiary (3°) alcohol	a tertiary carbon atom	R_3COH	2-methylpropan-2-ol

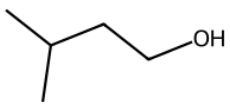
Q.6 [Short] Why is 2-methylpropan-2-ol a tertiary alcohol while propan-2-ol is a secondary alcohol?

In 2-methylpropan-2-ol, $(CH_3)_3C–OH$, the carbon bonded to the –OH group is directly attached to three other carbon atoms, so it is a tertiary carbon. In propan-2-ol, $(CH_3)_2CH–OH$, this carbon is attached to only two other carbon atoms, so it is a secondary carbon.

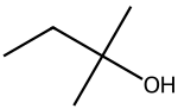
Quick Check 22.1

Q.7 [Short] I. Which ones of these are 1°, 2° and 3° alcohols?

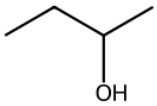
	Alcohol	–OH carbon is attached to	Type
a)	3-methylbutan-1-ol	one carbon atom	1°
b)	2-methylbutan-2-ol	three carbon atoms	3°
c)	butan-2-ol	two carbon atoms	2°
d)	pentan-1-ol	one carbon atom	1°
e)	3-methylbutan-2-ol	two carbon atoms	2°
f)	cyclopentanol	two ring carbon atoms	2°



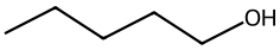
a) 3-methylbutan-1-ol



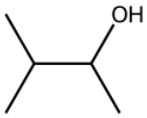
b) 2-methylbutan-2-ol



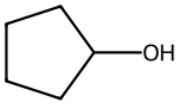
c) butan-2-ol



d) pentan-1-ol



e) 3-methylbutan-2-ol



f) cyclopentanol

22.2 ACIDITY OF ALCOHOLS AND PHENOLS

Q.8 [Long] **Why are alcohols weaker acids than water? Explain with equations.**

Alcohols have a low degree of dissociation: when dissolved in water, alcohol molecules do not dissociate to a great extent. The position of the equilibrium lies to the left, so there are far more alcohol molecules than RO^- and H^+ ions.

$$ROH_{(aq)} \rightleftharpoons RO^-_{(aq)} + H^+_{(aq)}$$
$$H_2O_{(l)} \rightleftharpoons OH^-_{(aq)} + H^+_{(aq)}$$

When water dissociates, the equilibrium also lies to the left, but there are more H^+ ions than in the dissociation of alcohols. As alcohols have a lower $[H^+]$ in solution, they are **weaker acids than water**.

Reason: alkyl groups are electron-donating groups (positive inductive effect). They push electrons towards the oxygen atom of the alkoxide ion, so there is more electron density on the O atom. The alkoxide ion is destabilized and is more likely to accept an H^+ ion and form the alcohol again.

Q.9 [Short] **What is an alkoxide ion? Why is it less stable than the hydroxide ion?**

The alkoxide ion, RO^-, is the ion left when an alcohol loses H^+. Its alkyl group is electron-donating and pushes electrons towards the oxygen atom, so there is more electron density on oxygen. This destabilises the ion, which readily accepts H^+ to form the alcohol again.

Q.10 [Long] Explain the acidity of phenol.

Although phenols contain an –OH group, they are weakly acidic. This is due to the delocalisation of one of the lone pairs of the oxygen atom into the aromatic ring: the lone pair overlaps with the delocalised π system of the ring. The delocalisation stabilises the **phenoxide ion**, the conjugate base of phenol: the negative charge is spread over the aromatic ring, so there is less charge density on the oxygen atom. The H^+ ions are therefore not strongly attracted to the phenoxide ion and are less likely to re-form the phenol molecule. So phenol is more likely to lose a proton (act as an acid) than to gain one.

$$C_6H_5OH_{(aq)} \rightleftharpoons C_6H_5O^-_{(aq)} + H^+_{(aq)}$$

Q.11 [Short] What is the phenoxide ion? Why is it stable?

The phenoxide ion, $C_6H_5O^-$, is the conjugate base of phenol. Its negative charge is delocalised (spread) over the aromatic ring through the overlap of an oxygen lone pair with the π system, so there is less charge density on the oxygen atom. This makes it stable.

Q.12 [Long] Compare the relative acidities of water, phenol and ethanol.

The acid strength depends on the stability of the conjugate base formed when the proton is lost.

	Ethanol	Water	Phenol
Conjugate base	ethoxide ion, $C_2H_5O^-$	hydroxide ion, OH^-	phenoxide ion, $C_6H_5O^-$
Effect on the negative charge	Ethyl group donates electrons (+I): more electron density on O; ion destabilized	No such effect	Charge delocalised over the benzene ring: ion stabilized
Acid strength	Weakest	Intermediate	Strongest of the three

Relative acidities: phenol > water > ethanol

Quick Check 22.2

Q.13 [Short] i. Compare the relative acidic strength of methanol and ethanol.

Methanol is slightly more acidic than ethanol. Alkyl groups donate electrons (positive inductive effect) and destabilize the alkoxide ion. The ethyl group donates more electron density than the methyl group, so the ethoxide ion is less stable than the methoxide ion and ethanol dissociates less.

Q.14 [Short] **ii. Showing the electron delocalisation explain why the phenoxide (PhO^-) is more stable than the ethoxide ion ($C_2H_5O^-$).**

In the phenoxide ion, a lone pair on the oxygen overlaps with the delocalised π system of the benzene ring. The negative charge is spread over the oxygen and the ring (extended π system), so the charge density on oxygen is low and the ion is stable. In the ethoxide ion there is no π system; the negative charge stays on the oxygen and is even increased by the electron-donating ethyl group, so the ion is less stable.

Q.15 [Short] **iii. Alkali metals react with stronger acids to produce the hydrogen gas. A piece of sodium is added to two beakers; one containing ethanol and the other phenol. a) Predict in which beaker the hydrogen gas will be evolved earlier and more rapidly. b) If the phenolphthalein indicator is already present before adding sodium to the above solutions, which solution will turn pink earlier?**

a) In the beaker containing **phenol**. Phenol is a stronger acid than ethanol, so it gives up its proton to sodium more readily and hydrogen is evolved earlier and more rapidly.

b) The pink colour appears when a base is formed. The **ethanol** solution is expected to turn pink earlier: the ethoxide ion formed is a strong base. In the phenol beaker the phenoxide ion formed is a much weaker base (it is stabilised by delocalisation), and the remaining phenol is itself acidic, so the solution becomes alkaline enough for the indicator only later.

Part b) is answered by reasoning from the relative base strengths of ethoxide and phenoxide; the textbook only states that sodium with ethanol and phenolphthalein gives the pink colour slowly, and with water immediately.

22.3 SYNTHESIS OF ALCOHOLS

Q.16 [Long] **Describe the methods of synthesis of alcohols with equations.**

Alcohols can be prepared from alkenes, halogenoalkanes, carbonyl compounds, carboxylic acids and esters.

i) Hydration of alkenes: hot steam reacts with an alkene, using concentrated phosphoric acid (H_3PO_4) as a catalyst; electrophilic addition takes place.

$$CH_2{=}CH_2 + H_2O_{(g)} \rightarrow CH_3{-}CH_2{-}OH\ (H_3PO_4,\ \text{heat})$$

ii) Oxidation of alkenes: with cold, dilute $KMnO_4$, a mild oxidising agent, a diol is formed.

$$CH_2{=}CH_2 + H_2O + [O] \rightarrow HO{-}CH_2{-}CH_2{-}OH\ (\text{ethane-1,2-diol})\ (KMnO_4,\ OH^-)$$

iii) From halogenoalkanes: the halogen atom is substituted by an –OH group on heating with aqueous NaOH (nucleophilic substitution).

$$CH_3{-}CH_2{-}Br + NaOH \rightarrow CH_3{-}CH_2{-}OH + NaBr\ (\text{heat})$$

iv) Reduction of aldehydes and ketones: with reducing agents such as $NaBH_4$ or $LiAlH_4$. Aldehydes give primary alcohols; ketones give secondary alcohols.

$$CH_3–CHO + 2[H] \rightarrow CH_3–CH_2–OH\ (LiAlH_4)$$
$$CH_3–CO–CH_3 + 2[H] \rightarrow CH_3–CH(OH)–CH_3\ (LiAlH_4)$$

v) Reduction of carboxylic acids: with $LiAlH_4$, to primary alcohols.

$$CH_3–COOH + 4[H] \rightarrow CH_3–CH_2–OH + H_2O\ (LiAlH_4)$$

vi) Hydrolysis of esters: when an ester is heated with dilute acid or alkali, a carboxylic acid and an alcohol are formed.

$$CH_3–COO–CH_2–CH_3 + H_2O \rightleftharpoons CH_3–COOH + CH_3–CH_2–OH\ (H^+_{(aq)})$$

Q.17 [Short] How is ethanol obtained from ethene? Name the catalyst and the type of reaction.

Hot steam is reacted with ethene using concentrated phosphoric acid (H_3PO_4) as catalyst. It is an electrophilic addition reaction.

$$CH_2=CH_2 + H_2O_{(g)} \rightarrow CH_3–CH_2–OH$$

Q.18 [Short] What is formed when ethene reacts with cold, dilute potassium manganate(VII)?

Cold dilute $KMnO_4$ is a mild oxidising agent. It oxidises ethene to a diol, ethane-1,2-diol, $HO–CH_2–CH_2–OH$.

Q.19 [Short] Which type of alcohol is formed by the reduction of (a) an aldehyde, (b) a ketone, (c) a carboxylic acid?

(a) An aldehyde gives a primary alcohol and (b) a ketone gives a secondary alcohol, with $NaBH_4$ or $LiAlH_4$. (c) A carboxylic acid gives a primary alcohol with $LiAlH_4$. For example, ethanal and ethanoic acid give ethanol; propanone gives propan-2-ol.

Q.20 [Short] What are the products of hydrolysis of ethyl ethanoate?

When the ester is heated with dilute acid or alkali, it gives a carboxylic acid (ethanoic acid) and an alcohol (ethanol).

$$CH_3–COO–CH_2–CH_3 + H_2O \rightleftharpoons CH_3–COOH + CH_3–CH_2–OH$$

22.4 REACTIONS OF ALCOHOLS

Q.21 [Short] Why is alcohol used as an alternative fuel to petrol? Write the equation for the combustion of ethanol.

Alcohols, when ignited, undergo complete combustion to form carbon dioxide and water. The reaction is highly exothermic; this is why alcohol is used as an alternative to petrol in some countries.

$$CH_3–CH_2–OH + 3O_2 \rightarrow 2CO_2 + 3H_2O\ \Delta H°_c = -1367\ kJ\ mol^{-1}$$

Q.22 [Long] Describe the substitution reactions of alcohols which give halogenoalkanes.

In the substitution of alcohols, the hydroxy group (–OH) is replaced by another substituent (e.g., a halogen) to form a new product (e.g., a halogenoalkane).

$$CH_3–CH_2–OH + HCl \rightarrow CH_3–CH_2–Cl + H_2O\ (ZnCl_2)$$
$$CH_3–CH_2–OH + SOCl_2 \rightarrow CH_3–CH_2–Cl + SO_2 + HCl$$
$$CH_3–CH_2–OH + PCl_5 \rightarrow CH_3–CH_2–Cl + POCl_3 + HCl$$
$$CH_3–CH_2–OH + KBr + H_2SO_4 \rightarrow CH_3–CH_2–Br + KHSO_4 + H_2O$$

In this chapter the textbook prints only the HCl/$ZnCl_2$ equation; the other equations are those given in Chapter 21 (section 21.1) and listed in the learning outcomes.

Quick Check 22.3

Q.23 [Short] a) Write chemical reaction of propan-1-ol with $SOCl_2$.

$$CH_3–CH_2–CH_2–OH + SOCl_2 \rightarrow CH_3–CH_2–CH_2–Cl + SO_2 + HCl$$

The product is 1-chloropropane; both by-products are gases.

Q.24 [Short] b) Differentiate the reaction of PCl_3 and PCl_5 with propan-1-ol with the help of the chemical equations.

$$3CH_3–CH_2–CH_2–OH + PCl_3 \rightarrow 3CH_3–CH_2–CH_2–Cl + H_3PO_3$$
$$CH_3–CH_2–CH_2–OH + PCl_5 \rightarrow CH_3–CH_2–CH_2–Cl + POCl_3 + HCl$$

With PCl_3, three moles of alcohol react with one mole of the reagent and phosphorous acid (H_3PO_3) is the by-product. With PCl_5, one mole of alcohol reacts with one mole of the reagent; the by-products are $POCl_3$ and steamy fumes of HCl.

Q.25 [Short] c) How ethanol reacts with KBr? give reaction equation.

Ethanol is heated with potassium bromide and concentrated sulphuric acid. The acid first reacts with KBr to give HBr, which then reacts with ethanol to give bromoethane.

$$CH_3–CH_2–OH + KBr + H_2SO_4 \rightarrow CH_3–CH_2–Br + KHSO_4 + H_2O$$

Q.26 [Long] Describe the reaction of alcohols with sodium. How is it used as a test for the hydroxy group?

When an alcohol reacts with a reactive metal such as sodium, the oxygen–hydrogen bond of the hydroxyl group breaks. The reaction is less vigorous than that of sodium with water; hydrogen gas is given off and a base (alkoxide) is formed.

$$2CH_3–CH_2–OH + 2Na \rightarrow 2CH_3–CH_2–ONa + H_{2(g)}$$

This reaction is used as a **test for the hydroxy (–OH) group**: bubbles of hydrogen gas are released if a hydroxy group is present. If the excess ethanol is evaporated off after

the reaction, a white crystalline solid of sodium alkoxide is left. The longer the hydrocarbon chain of the alcohol, the less vigorous the reaction.

Q.27 [Short] Why does phenolphthalein turn pink slowly when sodium is added to ethanol, but immediately with water?

Sodium reacts slowly with ethanol, so the base (sodium ethoxide) is formed slowly and the pink colour of phenolphthalein appears slowly. With water the reaction is vigorous and the base (sodium hydroxide) is formed at once, so the colour changes to pink immediately.

Q.28 [Short] How does the length of the carbon chain affect the reaction of an alcohol with sodium?

The longer the hydrocarbon chain of the alcohol, the less vigorous is the reaction with sodium.

Q.29 [Long] Describe the oxidation of primary, secondary and tertiary alcohols.

Primary alcohols can be oxidised to aldehydes (which can be further oxidised to carboxylic acids), and secondary alcohols can be oxidised only to ketones. Tertiary alcohols do not undergo oxidation. The oxidising agents are acidified $K_2Cr_2O_7$ and acidified $KMnO_4$.

$$CH_3–CH_2–OH + [O] \rightarrow CH_3–CHO + H_2O \text{ (ethanol → ethanal)}$$

$$CH_3–CH(OH)–CH_3 + [O] \rightarrow CH_3–CO–CH_3 + H_2O \text{ (propan-2-ol → propanone; heat)}$$

Acidified potassium dichromate(VI), $K_2Cr_2O_7$, is an orange oxidising agent. When alcohols are oxidised, the orange dichromate ions ($Cr_2O_7^{2-}$) are reduced to green Cr^{3+} ions. Only primary and secondary alcohols change the colour, so this is used to check for the presence of a primary or secondary alcohol: propan-1-ol and propan-2-ol turn the orange solution green, while 2-methylpropan-2-ol (tertiary) gives no colour change.

Alcohol	Product of oxidation	Acidified $K_2Cr_2O_7$
Primary	Aldehyde (by distillation); carboxylic acid (by refluxing)	Orange → green
Secondary	Ketone	Orange → green
Tertiary	Not oxidised	Stays orange

The text calls $K_2Cr_2O_7$ "potassium chromate(VI)" in one place; it is potassium dichromate(VI). Distillation/refluxing is stated in the learning outcomes.

Q.30 [Short] How can acidified potassium dichromate(VI) be used to distinguish a tertiary alcohol from primary and secondary alcohols?

Primary and secondary alcohols are oxidised, so the orange $Cr_2O_7^{2-}$ ions are reduced to green Cr^{3+} ions. A tertiary alcohol is not oxidised, so the solution stays orange.

Q.31 [Short] **Why must potassium dichromate(VI) be acidified to act as an oxidising agent?**

For potassium dichromate(VI) to act as an oxidising agent, it must itself be reduced. This reduction requires hydrogen (H^+) ions, which are provided by the acidic medium.

Q.32 [Short] **How can a primary alcohol be oxidised to (a) an aldehyde and (b) a carboxylic acid?**

(a) The alcohol is warmed with the acidified oxidising agent and the aldehyde is distilled off as soon as it forms. (b) The alcohol is refluxed with excess oxidising agent, so the aldehyde stays in the flask and is oxidised further to the carboxylic acid.

Q.33 [Short] **What is dehydration of alcohols? Name the dehydrating agents.**

Alcohols undergo dehydration to form alkenes. Dehydration is a type of elimination reaction in which a water molecule is removed. Dehydrating agents may be aluminium oxide, conc. sulphuric acid, conc. phosphoric acid, etc.

$$CH_3–CH_2–OH \rightarrow CH_2=CH_2 + H_2O \ (Al_2O_3, \text{heat})$$

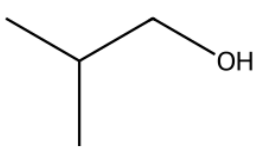
Q.34 [Short] **What is a breathalyser? On which reaction does it work?**

Driving after consuming alcohol is very dangerous. A breathalyser is a device used by traffic officers to check whether a driver is drunk or not. It uses the oxidation of ethanol, which requires electrons; the amount of current used is proportional to the amount of alcohol consumed.

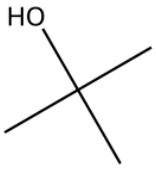
Quick Check 22.4

Q.35 [Short] **a) Three alcohols A, B and C are structural isomers of $C_4H_{10}O$. Each alcohol is refluxed with acidified dichromate(VI), $H^+/Cr_2O_7^{2-}$. i) Write the systematic name for alcohol C. ii) Write the equation for the complete combustion of alcohol A.**

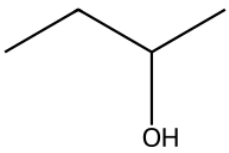
The three structures are shown below (i = A, ii = B, iii = C).



A: 2-methylpropan-1-ol



B: 2-methylpropan-2-ol



C: butan-2-ol

i) Alcohol C is **butan-2-ol**.

ii) Alcohol A is 2-methylpropan-1-ol:

$$C_4H_{10}O + 6O_2 \rightarrow 4CO_2 + 5H_2O$$

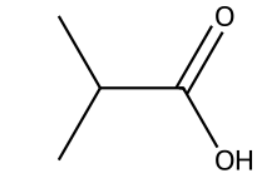
Q.36 [Short] b) Acidified potassium manganate(VII), $KMnO_4$, is a purple oxidising agent. What colour change would you observe when alcohols are oxidised with manganate ions (MnO_4^-)?

The purple colour of the MnO_4^- ions disappears: the solution turns from **purple to colourless**, as MnO_4^- is reduced to nearly colourless Mn^{2+} ions. (This happens with primary and secondary alcohols only.)

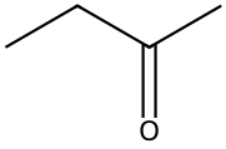
The ion is printed as "MnO^{4-}" in the textbook; the manganate(VII) ion is MnO_4^-.

Q.37 [Short] c) Draw the structures for the organic products of the oxidation of the following. If there is no reaction, write 'NONE'.

Alcohol	Type	Organic product (reflux with acidified dichromate)
A: 2-methylpropan-1-ol	Primary	2-methylpropanoic acid, $(CH_3)_2CH–COOH$ (via 2-methylpropanal)
B: 2-methylpropan-2-ol	Tertiary	NONE
C: butan-2-ol	Secondary	butanone, $CH_3–CO–CH_2–CH_3$



from A: 2-methylpropanoic acid



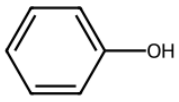
from C: butanone

No separate structures are printed for part c); it is answered for alcohols A, B and C of part a).

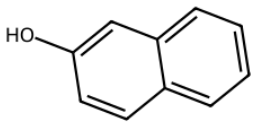
22.5 PHENOLS

Q.38 [Short] What are phenols? Give examples.

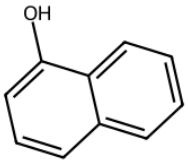
Compounds in which a hydroxyl group is bonded to an arene are called **phenols**. The simplest phenol has the formula C_6H_5OH. Phenol is also known as carbolic acid and was used as carbolic soap. It is a crystalline white solid. Other examples are 1-naphthol and 2-naphthol.



phenol



2-naphthol



1-naphthol

Q.39 [Short] Why is phenol called carbolic acid?

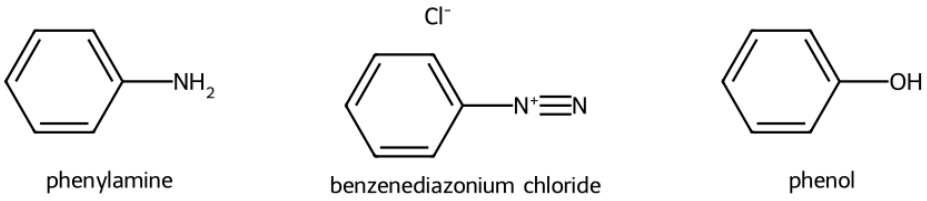
Phenol is weakly acidic although it contains an –OH group, so it is also known as carbolic acid; it was used in carbolic soap. It is a white crystalline solid.

Q.40 [Long] How is phenol prepared from phenylamine? Give reagents and conditions.

Phenols can be produced by the reaction of phenylamine with nitrous acid (HNO_2). The nitrous acid is produced *in situ*. The reaction occurs in three steps:

1. HNO_2 is so unstable that it is prepared in the test-tube by reacting sodium nitrite ($NaNO_2$) with dilute hydrochloric acid (HCl), keeping the temperature below 10 °C using ice.
2. Phenylamine reacts with the HNO_2 to form an unstable diazonium salt.
3. On further warming, the diazonium salt is decomposed (hydrolysed) and phenol is produced.

$$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$$
$$C_6H_5NH_2 + HNO_2 + HCl \rightarrow C_6H_5N_2^+Cl^- + 2H_2O \text{ (below 10 °C)}$$
$$C_6H_5N_2^+Cl^- + H_2O \rightarrow C_6H_5OH + N_2 + HCl \text{ (warm)}$$



phenylamine | benzenediazonium chloride | phenol

The textbook shows the sequence as a scheme ($NaNO_2$, HCl, < 10 °C; then warming, with NaOH written on the second arrow and NaCl + N_2 as by-products). The balanced equations above are the standard ones; the learning outcome says "warming of the diazonium salt with H_2O".

Q.41 [Short] Why is nitrous acid prepared in situ, and why is ice used?

HNO_2 is very unstable, so it cannot be stored; it is prepared in the test-tube itself by reacting $NaNO_2$ with dilute HCl. The temperature is kept below 10 °C with ice because the acid and the diazonium salt formed are unstable and decompose on warming.

22.6 REACTIONS OF PHENOL

Q.42 [Long] Describe the reactions of the –OH group of phenol.

The –OH group in phenols has a slightly acidic character, so it can act as an acid and take part in acid–base reactions.

i) With bases: phenols are only slightly soluble in water because of the large non-polar benzene ring, but they dissolve in alkaline solutions, forming a soluble salt and water.

$$C_6H_5OH + NaOH \rightarrow C_6H_5O^-Na^+ + H_2O \text{ (sodium phenoxide)}$$

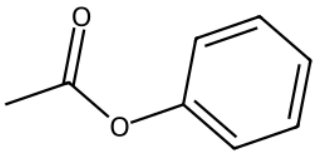
ii) With reactive metals: molten phenol reacts vigorously with sodium. This is also an acid–base reaction: a soluble salt is formed and hydrogen gas is given off.

$$2C_6H_5OH + 2Na \rightarrow 2C_6H_5O^-Na^+ + H_2$$

iii) With acyl chloride: when phenol is heated with an acyl chloride in the presence of a base, an ester is produced. The base deprotonates the phenol to form a phenoxide ion, which is a better nucleophile than phenol itself and is able to attack the carbonyl carbon.

$$CH_3COCl + C_6H_5OH \rightarrow CH_3COOC_6H_5 \text{ (phenyl ethanoate)} + HCl \text{ (NaOH, heat)}$$

iv) With diazonium ions: this reaction is discussed in Chapter 24 (Organic Nitrogen Compounds).



phenyl ethanoate

The textbook sentence says "an ether is produced"; the product it draws and names, phenyl ethanoate, is an ester.

Q.43 [Short] Why is phenol only slightly soluble in water but dissolves in sodium hydroxide solution?

The large non-polar benzene ring makes phenol only slightly soluble in water. With an alkali it reacts as an acid and forms sodium phenoxide, an ionic salt which is soluble in water.

Q.44 [Short] Why is a base used in the reaction of phenol with an acyl chloride?

The base removes the proton of phenol to form the phenoxide ion. The phenoxide ion is a better nucleophile than phenol and can attack the carbonyl carbon of the acyl chloride to form the ester.

Q.45 [Long] Why is the aromatic ring of phenol more reactive than benzene? To which positions are electrophiles directed?

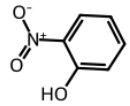
In phenol the oxygen atom of the hydroxyl group donates electron density into the ring: one of its lone pairs overlaps with the π system of the benzene ring and becomes delocalised. This increases the electron density in the aromatic ring, so the ring is more likely to undergo electrophilic attack; it is **activated**. The incoming electrophiles are directed by the hydroxyl group to the 2, 4 and 6 positions.

Q.46 [Long] Describe the nitration of phenol.

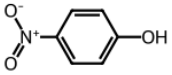
With dilute nitric acid at room temperature, phenol gives a mixture of 2-nitrophenol and 4-nitrophenol. With concentrated HNO_3 the product is 2,4,6-trinitrophenol. A hydrogen atom of the ring is substituted by a nitro ($-NO_2$) group.

$$2C_6H_5OH + 2HNO_3 \rightarrow 2\text{-}O_2NC_6H_4OH + 4\text{-}O_2NC_6H_4OH + 2H_2O \text{ (dilute } HNO_3\text{, room temperature)}$$

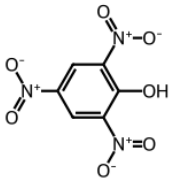
$$C_6H_5OH + 3HNO_3 \rightarrow C_6H_2(NO_2)_3OH + 3H_2O \text{ (conc. } HNO_3\text{)}$$



2-nitrophenol



4-nitrophenol

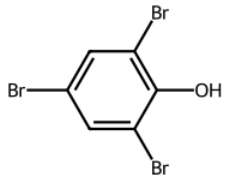


2,4,6-trinitrophenol

Q.47 [Short] Describe the reaction of phenol with bromine water.

Phenol reacts with bromine water at room temperature. It decolourises the orange bromine solution and forms a white precipitate of 2,4,6-tribromophenol.

$$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr \text{ (room temperature)}$$



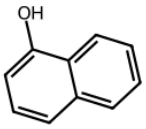
2,4,6-tribromophenol

Q.48 [Long] Compare the nitration and bromination of phenol with those of benzene.

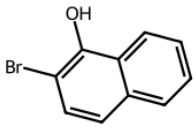
Reaction	Benzene	Phenol
Nitration	Mixture of conc. HNO_3 and conc. H_2SO_4, refluxed between 25 °C and 60 °C; gives nitrobenzene.	Dilute HNO_3 at room temperature; gives 2- and 4-nitrophenol. Conc. HNO_3 gives 2,4,6-trinitrophenol.
Bromination	Pure bromine (not a solution) and anhydrous $AlBr_3$ catalyst at room temperature; gives bromobenzene.	Bromine water at room temperature, no catalyst; gives a white precipitate of 2,4,6-tribromophenol.

Q.49 [Long] Describe the electrophilic substitution of 1-naphthol.

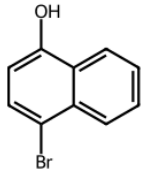
There are many phenolic compounds other than phenol; an example is 1-naphthol, which has a hydroxy group at position 1 of naphthalene. Like phenol, its –OH group is electron-donating and activates the naphthalene ring to electrophilic substitution. The electrophiles are directed to the 2 and/or 4 positions. Substitution at positions 9 and 10 is not possible, as there is no hydrogen atom on these carbons. 1-Naphthol and other phenolic compounds react in a similar way to phenol.



1-naphthol



2-bromo-1-naphthol



4-bromo-1-naphthol

1-Naphthol with Br_2 in CH_3COOH gives 2-bromo- and 4-bromo-1-naphthol

Q.50 [Short] **Why is substitution at positions 9 and 10 of 1-naphthol not possible?**

These are the two carbon atoms shared by both rings. They carry no hydrogen atom, so there is nothing to be substituted.

Quick Check 22.5

Q.51 [Short] **a) The reaction of phenol with bromine water is used for the identification of phenol. What will be the visual change during the reaction, also give the systematic name of the compound formed.**

The orange colour of bromine water is decolourised and a **white precipitate** is formed. The compound is **2,4,6-tribromophenol**.

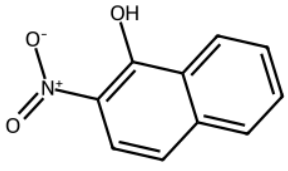
Q.52 [Short] **b) Nitration is an example of aromatic electrophilic substitution and its rate depends upon the group already present in the benzene ring. Out of benzene and phenol, which one is more easily nitrated and why?**

Phenol. A lone pair on the oxygen of the –OH group is delocalised into the ring, which increases the electron density of the ring and activates it towards electrophiles. So phenol is nitrated by dilute HNO_3 at room temperature, whereas benzene needs a mixture of conc. HNO_3 and conc. H_2SO_4.

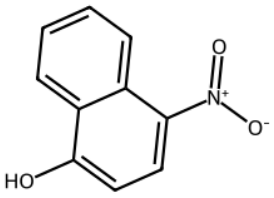
Q.53 [Short] **c) Write the reaction equation for nitration of 1-naphthol.**

The –OH group directs the nitro group to the 2 and 4 positions, so a mixture of 2-nitro-1-naphthol and 4-nitro-1-naphthol is formed (dilute HNO_3, room temperature):

$$2C_{10}H_7OH + 2HNO_3 \rightarrow 2\text{-}O_2NC_{10}H_6OH + 4\text{-}O_2NC_{10}H_6OH + 2H_2O$$



2-nitro-1-naphthol



4-nitro-1-naphthol